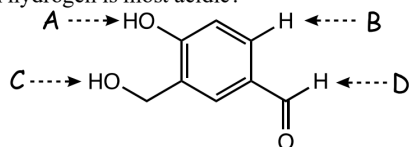
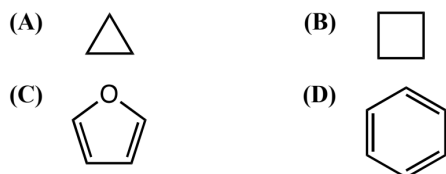


1. Which hydrogen is most acidic?

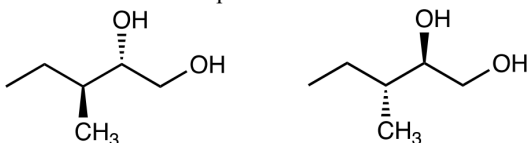


(A) A (B) B (C) C (D) D

2. In which compound are the carbon atoms not coplanar in the most stable conformation?



3. What is the relationship between the two structures?

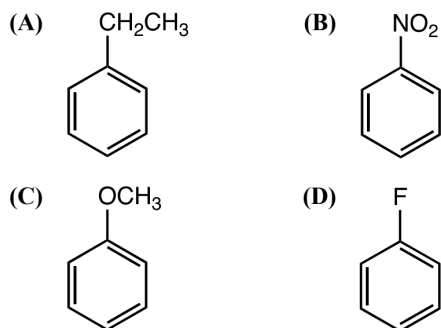


(A) Constitutional isomers (B) Diastereomers
(C) Enantiomers (D) Identical

4. What functional group is formed by the reaction of a Grignard reagent with an aldehyde?

(A) Ester (B) Ketone
(C) Secondary alcohol (D) Tertiary alcohol

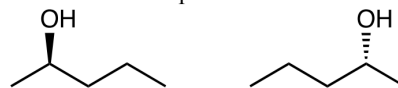
5. Which compound undergoes electrophilic aromatic substitution under the mildest conditions?



6. Which class of enzyme catalyzes the reaction of two molecules to form a larger product, usually using ATP as an energy source?

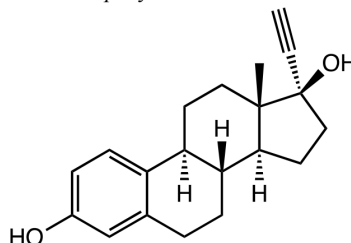
(A) Helicase (B) Ligase
(C) Protease (D) Racemase

7. What is the relationship between the two alcohols?



(A) Structural isomers (B) Enantiomers
(C) Diastereomers (D) Identical

8. In ethinylestradiol, shown below, how many sp^2 -hybridized and sp -hybridized carbons are there?



(A) 6 sp^2 , 2 sp (B) 5 sp^2 , 8 sp
(C) 11 sp^2 , 8 sp (D) 12 sp^2 , 8 sp

9. Which statement most accurately describes the most stable conformation of cyclohexane?

(A) All six carbon atoms are in the same plane.
(B) All bond angles are 109.5° .
(C) All hydrogens are eclipsed relative to the hydrogen atoms on adjoining carbons.
(D) The carbon-carbon bonds alternate between short and long distances.

10. The tertiary alcohol 2-methyl-2-propanol reacts with concentrated aqueous hydrochloric acid at 0°C to form 2-chloro-2-methylpropane. How is the mechanism of this reaction best described?

(A) S_N1 (B) S_N2 (C) E1 (D) E2

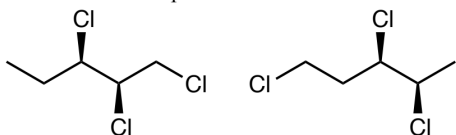
11. Which statement most accurately describes the properties of ketones and esters?

(A) Ketones are weak acids; esters are strong acids.
(B) Ketones react by nucleophilic addition; esters react by nucleophilic substitution.
(C) Ketones are readily reduced; esters are readily oxidized.
(D) The carbonyl group of a ketone is resonance stabilized; the carbonyl group of an ester is not resonance stabilized.

12. Which is not found in a typical nucleic acid?

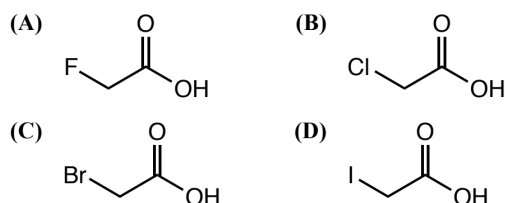
(A) Carboxylic acid (B) Dialkyl phosphate
(C) Heterocyclic base (D) Five-carbon sugar

13. What is the relationship between these two molecules?

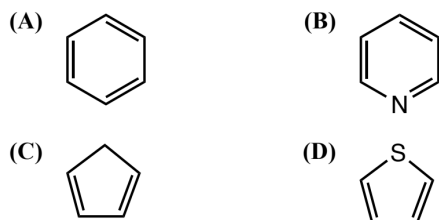


- (A) Identical
(B) Enantiomers
(C) Diastereomers
(D) Structural isomers

14. Which is the strongest acid?



15. Which molecule is not aromatic?



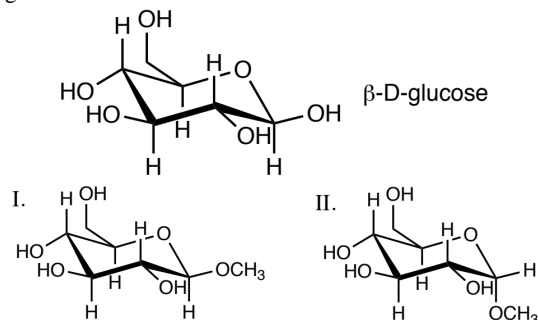
16. Which statement best describes the redox properties of alcohols?

- (A) Primary alcohols are oxidized to ketones by acidic dichromate.
(B) Ketones are reduced to primary alcohols by sodium borohydride.
(C) Secondary alcohols are oxidized to ketones by acidic dichromate.
(D) Tertiary alcohols are oxidized to esters by acidic dichromate.

17. Acetic acid reacts with ethanol in the presence of a strong acid catalyst to form ethyl acetate (the Fischer esterification reaction). What is the role of the acid catalyst?

- (A) It makes the carboxylic acid more electrophilic by protonating the carbonyl oxygen.
(B) It makes the ethanol more nucleophilic by protonating the hydroxyl oxygen.
(C) It converts the hydroxyl group of ethanol to a better leaving group.
(D) It shifts the equilibrium in the direction of the ester product.

18. Which are products of the acid-catalyzed reaction of β -D-glucose with methanol?



- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

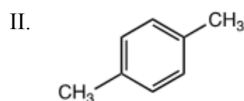
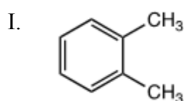
19. How many stable one-carbon molecules contain both hydrogen and chlorine and no other elements?

- (A) 2 (B) 3 (C) 4 (D) 5

20. Which property best accounts for the fact that acetic acid (CH_3COOH) is far more acidic than ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) in aqueous solution?

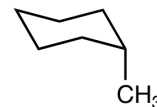
- (A) Acetic acid is more soluble in water than ethanol.
(B) Acetic acid has two hydrogen bond acceptors that can interact with water, while ethanol has only one.
(C) Acetic acid's conjugate base has its negative charge delocalized over two oxygens, while ethanol's conjugate base has its negative charge localized on one oxygen.
(D) Acetic acid can form a hydrate $\text{CH}_3\text{C}(\text{OH})_3$ in water, while ethanol cannot react with water in this way.

21. Which isomers of xylene would produce only a single mononitration product on treatment with $\text{HNO}_3/\text{H}_2\text{SO}_4$?

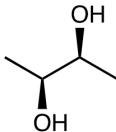
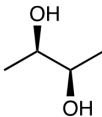
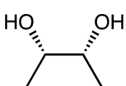
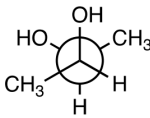
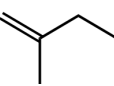
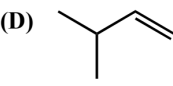
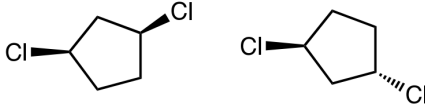
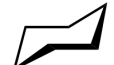
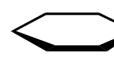


- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II
22. Under certain conditions, HBr adds to 2-methylpropene to give predominantly 2-bromo-2-methylpropane rather than isomeric 1-bromo-2-methylpropane. Which is the best mechanistic explanation for this regioselectivity?
-
- (A) H^+ adds preferentially to C-1 to form the more stable carbocation.
(B) Br^- adds preferentially to C-2 to form the more stable carbanion.
(C) $\text{H}^\bullet$ adds preferentially to C-1 to form the more stable radical.
(D) $\text{Br}^\bullet$ adds preferentially to C-2 to form the more stable radical.
23. Which will most reliably distinguish between an alcohol and an ether with the formula $\text{C}_4\text{H}_{10}\text{O}$?
- (A) The alcohol will have the higher normal boiling point.
(B) The alcohol will rapidly decolorize bromine while the ether will not.
(C) The alcohol will absorb visible light while the ether will not.
(D) The alcohol will form an electrically conductive aqueous solution while the ether will not.
24. Which form of the amino acid alanine is most abundant in aqueous solution at $\text{pH} = 7.0$?
- (A) $\text{CH}_3\text{CH}(\text{NH}_2)(\text{COOH})$
(B) $\text{CH}_3\text{CH}(\text{NH}_2)(\text{COO}^-)$
(C) $\text{CH}_3\text{CH}(\text{NH}_3^+)(\text{COOH})$
(D) $\text{CH}_3\text{CH}(\text{NH}_3^+)(\text{COO}^-)$
25. How many stable isomers have the formula $\text{C}_2\text{H}_2\text{ClBr}$?
- (A) 1 (B) 2 (C) 3 (D) 4

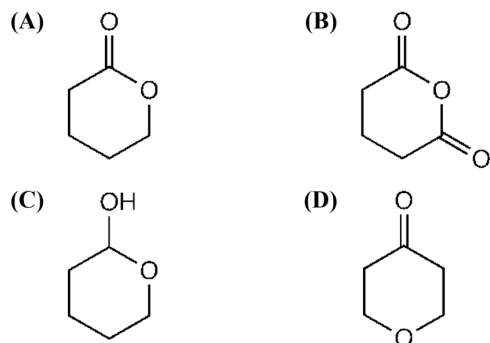
26. Which best describes the relationship between the two structures shown below?



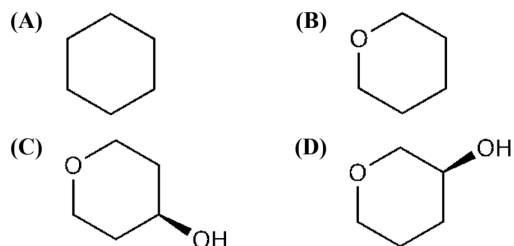
- (A) They are identical.
(B) They are enantiomers.
(C) They are in rapid equilibrium at room temperature and are present in equal quantities.
(D) They are in rapid equilibrium at room temperature and are present in unequal quantities.
27. Which compounds react readily with acidic dichromate solution ($\text{K}_2\text{Cr}_2\text{O}_7/\text{aqueous H}_2\text{SO}_4$)?
- I.
- II.
- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II
28. In the reaction of sodium cyanide with 1-bromobutane in methanol solution, what is the principal role of the cyanide ion?
- (A) A Brønsted base
(B) A nucleophile
(C) An oxidant
(D) A solvent
29. The amino acid lysine has the side chain $\text{R} = \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$. A mutation of lysine to which amino acid is least likely to change the overall charge on a protein at $\text{pH} = 8.0$?
- (A) Arginine, $\text{R} = \text{CH}_2\text{CH}_2\text{CH}_2\text{NHC}(=\text{NH})\text{NH}_2$
(B) Glycine, $\text{R} = \text{H}$
(C) Methionine, $\text{R} = \text{CH}_2\text{CH}_2\text{SCH}_3$
(D) Serine, $\text{R} = \text{CH}_2\text{OH}$
30. What monosaccharide units are found in both starch and cellulose?
- (A) Arabinose
(B) Fructose
(C) Glucose
(D) Ribose

31. A molecule has the formula $C_5H_{10}O$. Which functional groups might it contain?
- I. Alcohol
II. Ketone
- (A) I only (B) II only
(C) Either I or II (D) Neither I nor II
32. How many distinct branched alkanes are there with the formula C_6H_{14} ?
- (A) 1 (B) 2 (C) 3 (D) 4
33. Which cycloalkane in its most stable conformation has average internal C–C–C bond angles most different from the value given?
- (A) Cyclopropane, 60°
(B) Cyclobutane, 90°
(C) Cyclopentane, 108°
(D) Cyclohexane, 120°
34. Which gas is most reactive toward 1-hexene?
- (A) N_2 (B) O_3 (C) N_2O (D) NH_3
35. Which structure depicts the enantiomer of the diol shown?
- 
- (A) 
- (B) 
- (C) 
- (D) The diol shown is achiral and does not have an enantiomer.
36. Which biomolecule does NOT contain C=O double bonds?
- (A) Sucrose (a carbohydrate)
(B) Glyceryl trioleate (a lipid)
(C) Phenylalanine tRNA (a nucleic acid)
(D) Trypsin (a protein)
37. Which is NOT a valid representation of 3-methyl-1-butene?
- (A) $(CH_3)_2CHCH=CH_2$ (B) $CH_2CHCH(CH_3)_2$
(C)  (D) 
38. What is the relationship between the two compounds shown?
- 
- (A) Identical (B) Structural isomers
(C) Geometric isomers (D) Mirror image isomers
39. An amine with the formula $C_4H_{11}N$ will have which of these properties?
- (A) Unpleasant odor
(B) Boiling point greater than $100^\circ C$
(C) Absorption of light with $\lambda > 450\text{ nm}$
(D) Water solubility less than 10 g/L
40. Which combination of reactants and catalyst will produce methyl propanoate, $CH_3CH_2COOCH_3$, upon heating?
- (A) CH_3CH_2OH and CH_3COOH with catalytic NaOH
(B) CH_3CH_2OH and CH_3COOH with catalytic H_2SO_4
(C) CH_3OH and CH_3CH_2COOH with catalytic NaOH
(D) CH_3OH and CH_3CH_2COOH with catalytic H_2SO_4
41. Which conformation of cyclohexane (C_6H_{12}) is most stable?
- (A)  (B) 
(C)  (D) 
42. Which element is NOT present in DNA?
- (A) H (B) N (C) P (D) S

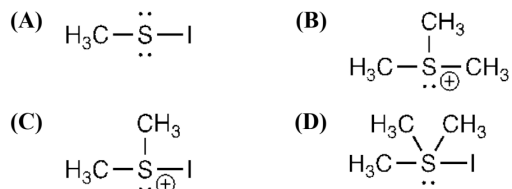
43. Which compound is an ester?



44. A pure substance is found to rotate the plane of plane-polarized light. Which compound is it?



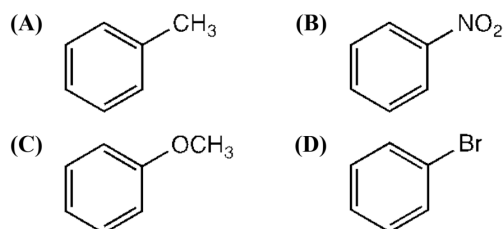
45. Which structure best represents the sulfur-containing species produced in the reaction of $(\text{CH}_3)_2\text{S}$ with CH_3I ?



46. How many distinct compounds have the formula C_5H_{12} ?

(A) One (B) Two (C) Three (D) Four

47. Which compound reacts the fastest in electrophilic aromatic substitution reactions (such as with Br_2 in the presence of FeBr_3)?



48. Which statement about unsaturated fats is NOT correct?

- (A) Unsaturated fats contain carbon-carbon double bonds.
- (B) Unsaturated fats generally have higher melting points than saturated fats with the same number of carbon atoms.
- (C) Unsaturated fats can be converted to saturated fats by treatment with H_2 in the presence of a metal catalyst.
- (D) Unsaturated fats must contain oxygen.

49. Which compound can exhibit geometric isomerism?

- (A) 1-butene (B) 2-butene
- (C) 1-butyne (D) 2-butyne

50. Which structural features contribute to the low strain energy of cyclohexane?

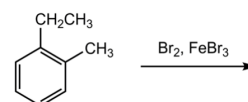
- I. All bond angles are close to 109.5° .
- II. All C-H bonds on adjacent carbon atoms are staggered.

- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

51. What reaction conditions most effectively convert a carboxylic acid to a methyl ester?

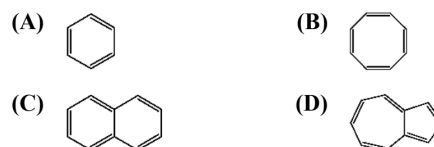
- (A) CH_3OH , HCl (B) CH_3I , HCl
- (C) CH_3OH , NaOH (D) CH_3I , SOCl_2

52. What is the maximum number of monosubstitution products of the aromatic substitution reaction shown?



- (A) 1 (B) 2 (C) 3 (D) 4

53. Which compound is NOT considered aromatic?



54. To what elements are the phosphorus atoms in DNA bonded?

- I. C
- II. H

- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

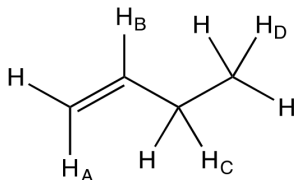
55. How many π bonds are in a molecule of propyne, C_3H_4 ?

- (A) 0 (B) 1 (C) 2 (D) 3

56. How many isomers have the formula $C_2H_2Br_2$?

- (A) 1 (B) 2 (C) 3 (D) 4

57. One hydrogen in 1-butene is replaced by bromine to give a chiral molecule. Which hydrogen is replaced?



- (A) H_A (B) H_B (C) H_C (D) H_D

58. A compound with the formula C_6H_{12} does not decolorize bromine in $CHCl_3$ solution. Which compound could this be?

- (A) Cyclohexane (B) 1-Hexene
(C) *Trans*-3-hexene (D) *Cis*-3-hexene

59. Which reaction would yield a single alkene product?

- (A)
$$\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_2\text{CH}_3 \end{array} \xrightarrow{50\% \text{H}_2\text{SO}_4} \text{C}_6\text{H}_{12}$$
- (B)
$$\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3\text{CH}_2\text{CCH}_2\text{CH}_3 \\ | \\ \text{CH}_2\text{CH}_3 \end{array} \xrightarrow{50\% \text{H}_2\text{SO}_4} \text{C}_7\text{H}_{14}$$
- (C)
$$\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_2\text{CH}_3 \end{array} \xrightarrow[\text{C}_2\text{H}_5\text{OH}]{\text{KOH}} \text{C}_6\text{H}_{12}$$
- (D)
$$\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CCH}_2\text{CH}_3 \\ | \\ \text{CH}_3 \end{array} \xrightarrow[\text{C}_2\text{H}_5\text{OH}]{\text{KOH}} \text{C}_7\text{H}_{14}$$

60. The net chemical reaction of photosynthesis is best described as

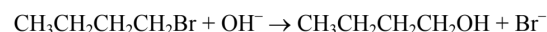
- (A) an endothermic reaction that forms sugars.
(B) an exothermic reaction that forms sugars.
(C) an endothermic reaction that breaks down sugars.
(D) an exothermic reaction that breaks down sugars.

61. What is the relationship between the following two molecules?



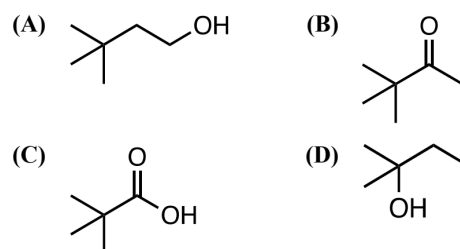
- (A) Structural isomers (B) Geometric isomers
(C) Enantiomers (D) Identical

62. What is the role of hydroxide ion in the reaction below?

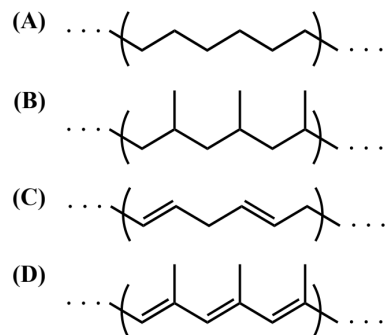


- (A) Oxidizing agent (B) Lewis acid
(C) Catalyst (D) Nucleophile

63. Which compound reacts readily with Cr(VI) reagents?



64. What is the structure of polypropylene, formed by the polymerization of propene ($\text{CH}_2=\text{CHCH}_3$)?



65. The sequence of amino acids in a protein is known as its

- (A) primary structure. (B) secondary structure.
(C) tertiary structure. (D) quaternary structure.

66. Molecules from which class of biopolymers can react with water, in the presence of suitable enzymes, to form smaller examples of that class of biopolymers?
- I. Proteins II. Polysaccharides
- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II
67. How many sigma bonds are in 2-butyne ($\text{CH}_3\text{C}\equiv\text{CCH}_3$)?
- (A) 3 (B) 6 (C) 9 (D) 11
68. Which alcohol will undergo acid-catalyzed dehydration under the mildest conditions?
- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- (C) $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ (D) $(\text{CH}_3)_3\text{COH}$
69. A 1.0 M aqueous solution of which compound has the lowest pH?
- (A) $\text{CH}_3\text{CH}_2\text{OH}$ (B) CH_3COOH
- (C) CH_3CHO (D) Cl_3CCHO
70. A six-carbon organic compound containing oxygen is suspected of being either a secondary alcohol or a ketone. Which chemical or physical test would best distinguish between these two possibilities?
- (A) Water solubility
- (B) Melting point
- (C) Treatment with sodium bicarbonate
- (D) Treatment with acidic dichromate
71. Which two bases are found as a hydrogen-bonded base pair in DNA?
- (A) A and T (B) C and T
- (C) C and U (D) G and U
72. What is the nature of the peptide bonds in a protein?
- (A) Hydrogen bonds (B) Amide bonds
73. Hydrogenation of an alkene converts it to an
- (A) alkane. (B) alkyne.
- (C) alcohol. (D) aldehyde.
74. What is the relationship between the two compounds below?
- $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_3$
- (A) Identical (B) Stereoisomers
- (C) Geometric isomers (D) Structural isomers
75. Which class of organic compounds does NOT contain oxygen?
- (A) Alcohol (B) Amide
- (C) Amine (D) Ketone
76. Compared to ionic compounds of similar molar mass, hydrocarbons typically have
- I. Higher water solubility II. Higher melting points
- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II
77. Which of the following compounds could contain exactly one triple bond?
- (A) C_5H_{10} (B) C_5H_{12} (C) C_6H_{10} (D) C_6H_{12}
78. Which of the following is NOT classified as a biopolymer?
- (A) Collagen (B) Glucose
- (C) Cellulose (D) Chitin
79. Which of the following statements regarding structural isomers is correct? They have
- (A) the same molecular formula but the atoms are bonded in different ways.
- (B) the same molar mass but different atoms in their composition.
- (C) the same physical properties but different chemical properties.
- (D) different isotopes in their structures.
80. What is the IUPAC name for the compound $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$?
- (A) pentanal (B) pentanoic acid
- (C) 1-pentanol (D) 1-pentanone
81. Which compound will react rapidly with Br_2 ?
- (A) benzene (B) hexane
- (C) 1-hexene (D) 1-hexanol
82. The reaction below is best classified as
- $$\text{CH}_3\text{OH} + \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow{\text{H}^+} \text{HC}-\overset{\text{O}}{\parallel}{\text{OCH}_3} + \text{H}_2\text{O}$$
- (A) addition. (B) esterification.
- (C) neutralization. (D) saponification.

Organic/Biochemistry

83. Which element is not found in common amino acids?

- (A) hydrogen (B) nitrogen
(C) oxygen (D) phosphorus

84. Which of the following is not classified as a carbohydrate?

- (A) glucose (B) glycine
(C) lactose (D) starch

85. What is the number of structural isomers with the molecular formula C_6H_{14} ?

- (A) three (B) four (C) five (D) six

86. All of the following are condensation polymers except

- (A) nylon (B) polyethylene
(C) protein (D) starch

87. Methanol can be gently oxidized with hot copper metal. What is(are) the product(s) of this oxidation?

- (A) acetic acid (B) carbon dioxide + water
(C) ethanol (D) methanal

88. Which statement does not describe benzene, C_6H_6 ?

- (A) It is an aromatic hydrocarbon.
(B) It exists in two isomeric forms.
(C) It undergoes substitution reactions.
(D) It can react to form three different products with the formula $C_6H_4Cl_2$.

89. All of the following atoms comprise part of a peptide functional group except

- (A) hydrogen. (B) nitrogen.
(C) oxygen. (D) phosphorus.

90. Which vitamin is the most soluble in water?

- (A) A (B) K (C) C (D) D

91. Which molecule contains the fewest hydrogen atoms?

- (A) cyclopropane (B) propane
(C) propene (D) propyne

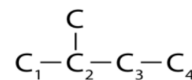
92. What is the difference between 2-chloropentane and 3-chloropentane?

- (A) the number of carbon atoms in the molecule
(B) the number of chlorine atoms in the molecule
(C) the position of the chlorine atom in the molecule
(D) the geometry of the carbon chain

93. Alanine is a(n)

- (A) amino acid. (B) ester.
(C) nucleic acid. (D) sugar.

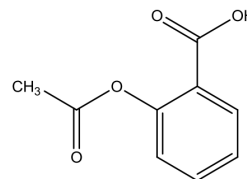
94. A secondary alcohol results from the attachment of a hydroxyl group to which carbon atom?



- (A) C_1 (B) C_2 (C) C_3 (D) C_4

95. Aspirin has the structural formula below. Which functional groups does this molecule contain?

- I acid II alcohol III ester IV ether



- (A) I and III (B) II and III
(C) II and IV (D) I, II, III and IV

96. Which substance yields the most energy per gram of sample upon metabolism?

- (A) carbohydrate (B) fat
(C) protein (D) vitamin

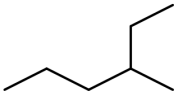
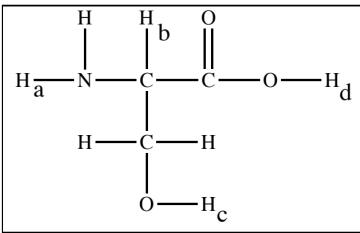
97. The boiling points of CH_3COCH_3 , $CH_3COC_2H_5$, and $CH_3COC_3H_7$ are $56^\circ C$, $80^\circ C$, and $102^\circ C$, respectively. This increase is best attributed to an increase in which of the following?

- I dipole-dipole interactions
II dispersion forces
III hydrogen bonding

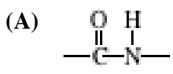
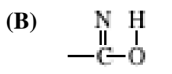
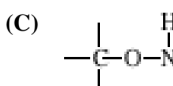
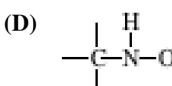
- (A) I only (B) II only
(C) III only (D) II and III only

98. How many isomeric compounds have the formula C_3H_8O ?

- (A) 1 (B) 2 (C) 3 (D) 4

99. How many sigma bonds does propyne, C_3H_4 , have?
 (A) 2 (B) 4 (C) 6 (D) 8
100. Which of the compounds below will react readily with chlorine by addition?
 I C_2H_2 (ethyne) II C_2H_4 (ethene) III C_6H_6 (benzene)
 (A) I only (B) I and II only
 (C) II and III only (D) I, II, and III
101. The smell of rancid butter is due to the presence of butyric (butanoic) acid. What is the formula for this compound?
 (A) $C_4H_8O_2$ (B) $C_4H_{10}O$
 (C) $C_5H_{10}O_2$ (D) $C_5H_{12}O$
102. In a double helix of DNA, adenine is paired with
 (A) adenine. (B) cytosine.
 (C) guanine. (D) thymine.
103. What is the IUPAC name for the molecule below?

 (A) heptane (B) 2-ethylpentane
 (C) 3-methylhexane (D) 4-ethylpentane
104. How many different aldehydes have the formula $C_5H_{10}O$?
 (A) 2 (B) 3 (C) 4 (D) 5
105. Which compound will rapidly decolorize bromine in $CHCl_3$?
 (A) benzene (B) cyclohexane
 (C) hexane (D) 1-hexene
106. Which combination of reactants produces an ester?
 (A) acid and alcohol (B) acid and aldehyde
 (C) alcohol and aldehyde (D) aldehyde and potassium
107. Which element is used to form cross-links between the strands of latex rubber?
 (A) Fe (B) N (C) P (D) S
108. The classification of a fat as saturated or unsaturated is based on whether
 (A) it can be metabolized by humans.
 (B) it contains carbon – carbon double bonds.
 (C) it has twenty or more carbon atoms.
 (D) it is of animal origin.
109. The formula, H_3CCOCH_3 , represents a(n)
 (A) aldehyde. (B) ester.
 (C) ether. (D) ketone.
110. Which suffix is used to designate a carbohydrate?
 (A) –ase (B) –ate (C) –one (D) –ose
111. Which compound has the largest molar mass?
 (A) hexane (B) 1-hexene
 (C) 1-hexyne (D) benzene
112. Which functional group is not commonly found in proteins?
 (A) alcohol (B) aldehyde
 (C) amide (D) amine
113. Which hydrogen is the most acidic in the molecule shown?

 (A) H_a (B) H_b (C) H_c (D) H_d
114. The gentle oxidation of ethanol, CH_3CH_2OH , produces
 (A) ethanal, CH_3CHO .
 (B) ethanoic acid, CH_3CO_2H .
 (C) carbon monoxide, CO .
 (D) carbon dioxide, CO_2 .
115. How many sigma bonds are in a molecule of diethyl ether, $C_2H_5OC_2H_5$?
 (A) 14 (B) 12 (C) 8 (D) 4

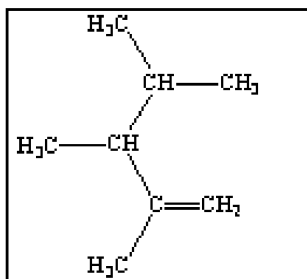
Organic/Biochemistry

116. Which term describes the formation of acetic acid from ethyl alcohol?
(A) addition (B) esterification
(C) neutralization (D) oxidation
117. How many structural isomers have the formula C_5H_{12} ?
(A) 2 (B) 3 (C) 4 (D) 5
118. Which class of compounds does not include C=O double bonds in its molecules?
(A) esters (B) amides
(C) alcohols (D) acids
119. What is the hybridization of the carbon atoms of C_2H_4 (ethene)?
(A) sp (B) sp^2 (C) sp^3 (D) s^2p^2
120. Which structure represents a peptide bond?
(A)  (B) 
(C)  (D) 
121. What is the molecular formula for a saturated compound named 2,2,4-trimethylpentane?
(A) C_7H_{14} (B) C_8H_{14} (C) C_8H_{16} (D) C_8H_{18}
122. How many different compounds have the formula C_3H_8O ?
(A) one (B) two (C) three (D) four
123. Which functional group does NOT contain an oxygen?
(A) alcohol (B) aldehyde
(C) amide (D) amine
124. How many pi bonds are present in a molecule of 1-butyne?
(A) one (B) two (C) three (D) four
125. Which is NOT an example of an addition polymer?
(A) polyethylene (B) polyethylene terephthalate
(C) polystyrene (D) polyvinyl chloride
126. The conversion of glucose to ethanol is represented:
$$x C_6H_{12}O_6 \rightarrow y C_2H_5OH + z CO_2$$

What are the coefficients x, y, z, respectively, in the balanced equation?
(A) 1, 2, 2 (B) 1, 3, 3 (C) 1, 1, 4 (D) 2, 4, 2
127. Which is an example of an aromatic compound?
(A) acetylene (B) benzene
(C) polyethylene (D) propanone
128. Which can exist as geometric isomers?
(A) 1,1-dichloroethane (B) 1,1-dichloroethene
(C) 1,2-dichloroethane (D) 1,2-dichloroethene
129. Which substance is formed when $K_2Cr_2O_7$ is added to C_2H_5OH in acid solution?
(A) C_2H_6 (B) CH_3COOH
(C) $K^+C_2H_5O^-$ (D) $C_2H_5OC_2H_5$
130. Which functional group, present in simple sugars, is responsible for their high solubility in water?
(A) -OH (B) -COOH
(C) -CONH₂ (D) -NH₂
131. Which family of compounds is used most frequently as flavoring agents?
(A) acids (B) alkenes
(C) esters (D) ethers
132. In addition to carbon, hydrogen and oxygen, which element is found in all amino acids?
(A) chlorine (B) nitrogen
(C) phosphorus (D) sulfur
133. Which is an isomer of 1-butanol?
(A) 1-propanol (B) butanone
(C) 1-chlorobutane (D) diethyl ether
134. Which compound is an aldehyde?
(A) CH_3OCH_3 (B) CH_3CHO
(C) CH_3COOH (D) CH_3COOCH_3
135. How many isomers exist for dibromobenzene ($C_6H_4Br_2$)?
(A) one (B) two (C) three (D) four

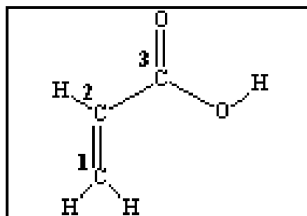
Organic/Biochemistry

136. What is the name for the compound with the structure?



- (A) 2-isopropyl-1-butene
(B) 2,3-dimethyl-2-hexene
(C) 2-methyl-3-isopropyl-1-butene
(D) 2,3,4-trimethyl-1-pentene

137. What is the hybridization of carbon atoms 1, 2, and 3, respectively in the structure?



- (A) sp^3 , sp , sp^2
(B) sp^2 , sp , sp^2
(C) sp^3 , sp^2 , sp^2
(D) sp^2 , sp^2 , sp^2

138. When egg white is coagulated the protein is said to be

- (A) condensed. (B) denatured.
(C) hydrolyzed. (D) polymerized.

139. Which formula represents n-butane?

- (A) $CH_3CH_2CH_2CH_3$ (B) $CH_2=CHCH_2CH_3$
(C) $(CH_3)_2CHCH_3$ (D) $(CH_3)_3CH$

140. How many structural isomers have the formula $C_3H_6Cl_2$?

- (A) 1 (B) 2 (C) 3 (D) 4

141. What is the hybridization of the carbon atom in a carboxyl group?

- (A) sp (B) sp^2 (C) sp^3 (D) dsp^3

142. A reaction in which a carboxylic acid reacts with an alcohol to form an organic compound and water is called

- (A) esterification (B) hydrolysis
(C) neutralization (D) saponification

143. What substance is formed when $CF_2=CF_2$ is polymerized?

- (A) Polyethylene (B) Polyurethane
(C) PVC (D) Teflon

144. Most enzymes are a type of

- (A) carbohydrate (B) lipid
(C) nucleic acid (D) protein

145. Which formula can be used to represent an alkynes?

- (A) C_nH_{2n-2} (B) C_nH_{2n}
(C) C_nH_{2n+2} (D) C_nH_{2n+4}

146. How many different structural isomers exist for dichloropropane, $C_3H_6Cl_2$?

- (A) 4 (B) 5 (C) 6
(D) some other number

147. All of the formulas below correspond to stable compounds EXCEPT

- (A) CH_2O (B) CH_2O_2
(C) CH_3O (D) CH_4O

148. Which of the compounds shown are isomers?

- 1 $CH_3CH_2OCH_3$
2 $CH_3CH_2OCH_2CH_3$
3 $CH_3CH_2CH_2OH$
4 $CH_2=CHOCH_3$

- (A) 1 and 3 (B) 1 and 2
(C) 2 and 3 (D) 1 and 4

149. Which functional group is present in CH_3COOH ?

- (A) aldehyde (B) carboxylic acid
(C) alcohol (D) hydroperoxide

150. How many sigma bonds does a molecule of ethene have?

- (A) 1 (B) 4 (C) 5 (D) 7

151. What is the oxidation product of a primary alcohol?

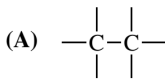
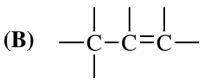
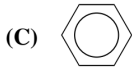
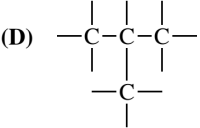
- (A) aldehyde (B) alkene
(C) ester (D) ketone

152. How many hydrogen atoms are in one molecule of propene?

- (A) 3 (B) 4 (C) 6 (D) 8

153. How many different compounds have the formula C_5H_{12} ?

- (A) 2 (B) 3 (C) 4 (D) 5

154. What bonds are present in $\text{H}-\text{C}\equiv\text{C}-\text{H}$?
 (A) 5 sigma (B) 4 sigma and 1 pi
 (C) 2 sigma and 3 pi (D) 3 sigma and 2 pi
155. In addition to carbon, hydrogen and oxygen, what else is found in every amino acid?
 (A) N (B) P
 (C) N and P (D) N and S
156. Which of these compounds contains a carboxyl group?
 (A) propanol (B) propanal
 (C) propanone (D) propanoic acid
157. Which class of compounds consists exclusively of saturated hydrocarbons?
 (A) alkanes (B) alkenes
 (C) alkynes (D) aromatics
158. How many different alcohols (not including optical isomers) have the molecular formula $\text{C}_4\text{H}_{10}\text{O}$?
 (A) 2 (B) 3 (C) 4 (D) 5
159. What is the position of the bromine atom relative to the methyl group in 3-bromotoluene?
 (A) meta (B) ortho
 (C) para (D) trans
160. Which of these compounds have *cis*- and *trans*- isomers?
 (A) dichlorobenzene (B) 1-chloropropene
 (C) 1,2-dichloropropane (D) dichloroethyne
161. If a certain polymer has the formula $(-\text{CH}_2\text{CCl}_2\text{CH}_2\text{CCl}_2-)_n$, from which monomer is it made?
 (A) $\text{HC}\equiv\text{CCl}$ (B) $\text{ClHC}=\text{CClH}$
 (C) $\text{Cl}_2\text{C}=\text{CH}_2$ (D) $\text{H}_2\text{C}=\text{CClH}$
162. How many isomers have the molecular formula C_5H_{12} ?
 (A) 1 (B) 2 (C) 3 (D) 5
163. Carbon is found in the highest oxidation state in which of these classes of organic compounds?
 (A) carboxylic acids (B) alcohols
 (C) aldehydes (D) alkynes
164. Which structural formula represents a mono-unsaturated aliphatic hydrocarbon?
 (A)  (B) 
 (C)  (D) 
165. Which compound has the highest boiling point?
 (A) $\text{CH}_3\text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{OCH}_2\text{CH}_3$
 (C) $\text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$